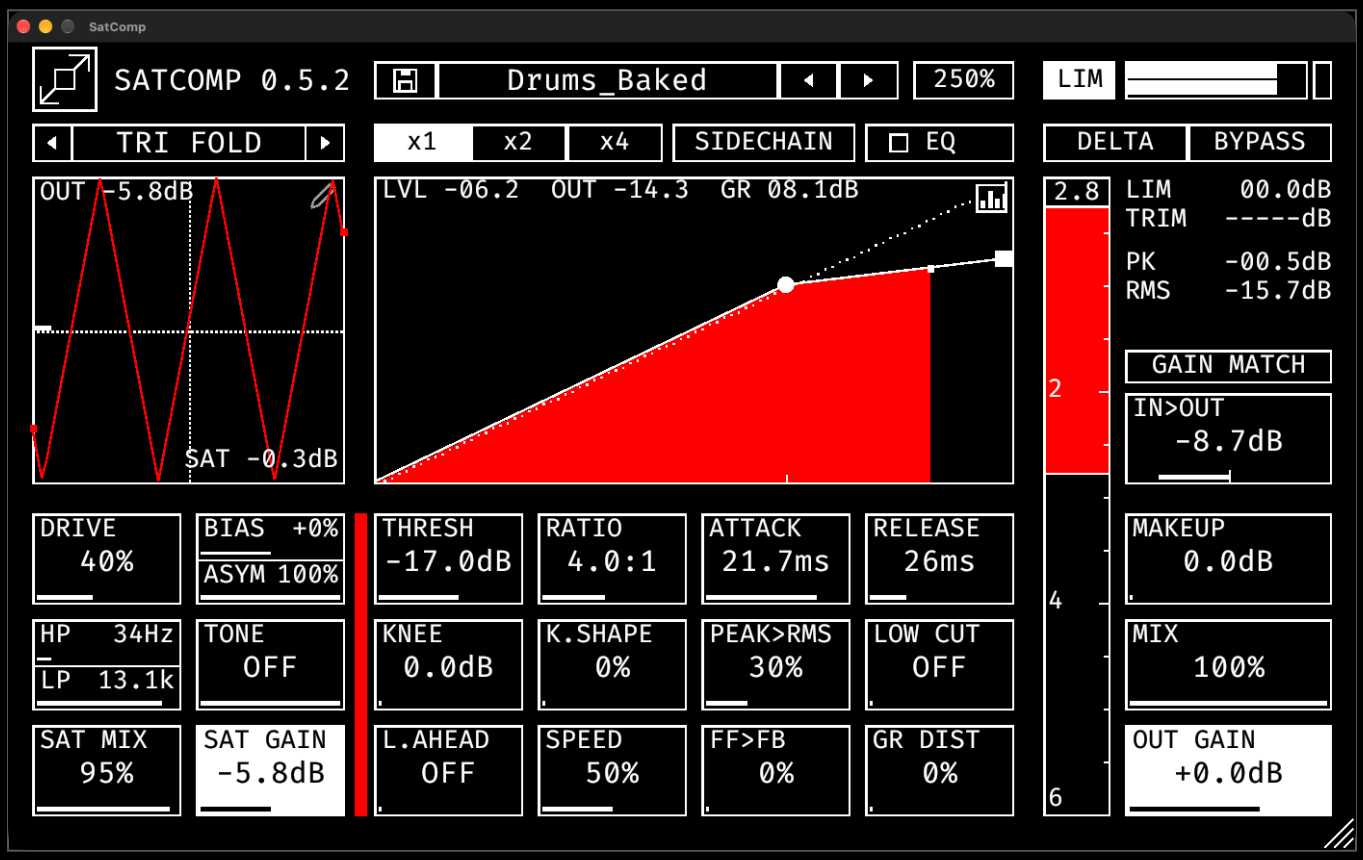
SatComp 0.5.2

waveshaping saturator with compressor and 2-stage 3-band EQ

A waveshaping saturator with a compressor and a 2-stage 3-band EQ. Give the compressor more spice to squash. Made with bricking intentions.



1. The idea

SatComp is a single strip: a saturator that feeds a compressor, an EQ that can sit before or after it, and a true-peak limiter at the end.

```
IN > [pre EQ] > SAT > COMP > MIX > [post EQ] > OUT GAIN > LIM > OUT
```

The saturator comes first on purpose: driving the shaper changes what the compressor hears, so the harder you saturate, the more there is to squash.

2. Layout

Area	What it does
Header	logo (themes), preset bar, window scale, oversampling, SIDECHAIN, EQ page, DELTA, BYPASS
Saturator (left)	shaper, DRIVE, BIAS, ASYM, HP, LP, TONE, SAT MIX, SAT GAIN and the transfer plot

Area	What it does
Compressor plot (middle)	transfer curve with the live operating point, the threshold dot and the ratio square
Compressor cells	THRESH, RATIO, ATTACK, RELEASE, KNEE, K.SHAPE, PEAK>RMS, LOW CUT, L.AHEAD, SPEED, FF>FB, GR DIST
Output column	LIM / TRIM / PK / RMS readouts, GAIN MATCH, IN>OUT, MAKEUP, MIX, OUT GAIN
Meters	post-saturator level meter, gain reduction meter, output meter

3. Saturator

Control	What it does
Shaper	15 waveshapers: Tube, Soft, Rational, Atan, Hard, Tri Fold, Sine Fold, Crossover, Cheby, Wrap, Rectify, Bitcrush, Asym Fold, Diode and **Custom**
DRIVE	how hard the signal is pushed into the shaper
BIAS	shifts the signal off-centre before the shaper: more even harmonics
ASYM	how much of the shaper's even-order part is kept. Lowering it stops lopsided curves from dragging DC and sub-bass pulses into the signal. Stored per shaper.
HP / LP	decide which part of the signal is saturated; the rest passes clean
TONE	low-pass on the shaped signal only
SAT MIX	blend between clean and saturated
SAT GAIN	level after the saturator, which is also the compressor's input level. Click the **SAT** readout in the plot corner to auto-match it.
x1 / x2 / x4	oversampling of the saturator (less aliasing, more latency)

Custom curve. Click the pencil icon in the plot corner to draw your own curve (up to 10 points). The symmetry tool draws one half and mirrors the other.

4. Compressor

Control	What it does
THRESH / RATIO	where compression starts and how much. You can also drag the dot (threshold) and the square (ratio) in the plot.
ATTACK / RELEASE	how fast the gain reacts
KNEE / K.SHAPE	width and curve of the transition around the threshold
PEAK>RMS	detector type: 0 % peak, 100 % RMS
LOW CUT	high-pass in the detector only, so kicks and bass stop driving the reduction. The audio is untouched.
L.AHEAD	lookahead: the detector sees the signal before the gain is applied, so the compressor is already down when a transient arrives
SPEED	scales attack and release together
FF>FB	detector topology: 0 % feed-forward, 100 % feedback
GR DIST	soft saturation after the gain reduction, growing with how hard the compressor works

> Sidechain

SIDECHAIN keys the compressor from the second stereo input, with key gain, HP and LP and a **SOLO** to listen to the key. **KEY MIX** takes the place of FF>FB and blends the key with the signal's own level. Routing belongs to the track, so the switch is saved with the project but not in presets.

5. EQ

The **EQ** button opens two EQ stages, one before and one after the compressor. Each has three bands plus a tilt, with a live spectrum. Use the pre stage to change what the compressor reacts to, the post stage to refine the tone.

6. Output

Control	What it does
MAKEUP	gain after the compressor
MIX	parallel blend of the processed signal with the dry input
OUT GAIN	final level
LIM	true-peak brickwall output limiter at -0.2 dBTP, on by default. Its reduction is the LIM readout.
GAIN MATCH	measures the loudness change for 3 seconds and stores a trim so processed and dry levels match
IN>OUT	net loudness change across the plug-in
DELTA	listen to what the compressor removes
BYPASS	plain bypass at the plug-in's latency: your dry signal at its own level

The **gain reduction meter** on the right shows the compressor's work. **Click** it to cycle its range. The number at the top is the reduction. The thin meter to its left is the level after the saturator; it turns accent for a second when the saturator output passes 0 dBFS. **Click** the output meter's top block to reset the clip lamp and the peak readout.

Waveform scope. Click the icon in the corner of the compressor plot for a half or full wave view of the output with the gain reduction stacked on top (a transient meter). Drag it to change the time window and the ceiling.

7. Recipes

Thick drums. Pick Tube or Diode, drive to 30-40 %, set ASYM to around 50 %, then compress with a fast attack and a medium release. Let the compressor squash what the saturator added.

Grit that follows the level. Raise GR DIST. The distortion grows with how hard the compressor works and vanishes when it is not reducing.

Ducking. Turn SIDECHAIN on and key from a kick. Raise LOW CUT to keep bass from fighting the key.

8. Good to know

- **Latency** is constant (about 2.2 ms at 48 kHz) and reported to the host, whether the limiter is on or off.

- **Presets:** six Drums presets ship with the plug-in; yours are saved to `SatComp/Presets` in your application-support folder. A name in asterisks means the preset has been modified.
- **Themes:** click the logo for the next scheme, double-click it for dark/light, right-click for the list. The title opens About, Preferences and the changelog.
- **About page:** click the title. **OPEN MANUAL (PDF)** opens this manual. The signal flow is live, and its PRE EQ, POST EQ and LIM blocks are switches you can click to bypass them.
- Double-click any slider to reset it.

Support and links

- Source code, issues and releases: github.com/thomchantra/SatComp
- Support the project: ko-fi.com/thomchantra

Found a bug or have an idea? Open an issue on GitHub. Thank you for using SatComp.